

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0671

Jones Chapel 500 kV
March 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0671. The Interconnection Customer proposes a 514.1 MW natural gas combined cycle facility interconnecting at the Jones Chapel 500 kV substation in the Indiana area. The request was received on September 01, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 271 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$7,175,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Scott – Shawville Sta 345.0 kV
Contingency: Orchard Rd – Shawville Sta 500.0 kV
Loading: 100.9% (Rating: 1,401 MVA)
DFAX: 5.1%
Pre-project loading: 99.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Scott – Shawville Sta 345.0 kV	Reconductor line	\$48,282,000	15% (\$7,175,000)
	Total allocated to project		\$7,175,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.